Multi-objective Optimization of Olive-Tree Pruning Biorefinery Networks Integrating Economic Performance and Life Cycle Environmental Impacts

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INTRODUCTION

Transitioning to sustainable bio-economies requires replacing fossil resources by mobilizing agricultural residues within circular supply chains. In Andalusia, **olive tree pruning generates 6.66 Mt year^{-1} of biomass**, an abundant resource that remains underutilized or restricted to low-value energy applications. However, designing regional biorefinery clusters is a complex **decision-making problem**, as **cost-effective configurations** do not necessarily **minimize life-cycle environmental burdens**. To address this challenge, this work presents a superstructure optimization framework to identify the optimal biomass allocation, technology deployment, and product distribution for **olive-pruning valorization networks in Andalusia**.

Keywords: Olive tree pruning • Multi-objective optimization • Life cycle assessment • Biomass Network • Decision-Making



METHODOLOGY

1. Superstructure Optimization:

-Models a network comprising **31 technologies** and **21 products**, including bio-based, fossil-reference and CCS-integrated pathways.

-**Required final products**: biochar, hydrogen, electricity, heat, ethanol, xylitol and antioxidants.

-LP model implemented in **Pyomo (Python)** using the **COIN-OR Branch & Cut** solver.

2. Multi-Objective Approach:

-Applies the **ε-constraint method** to **minimize** total annual system cost and life-cycle greenhouse gas (GHG) emissions simultaneously.

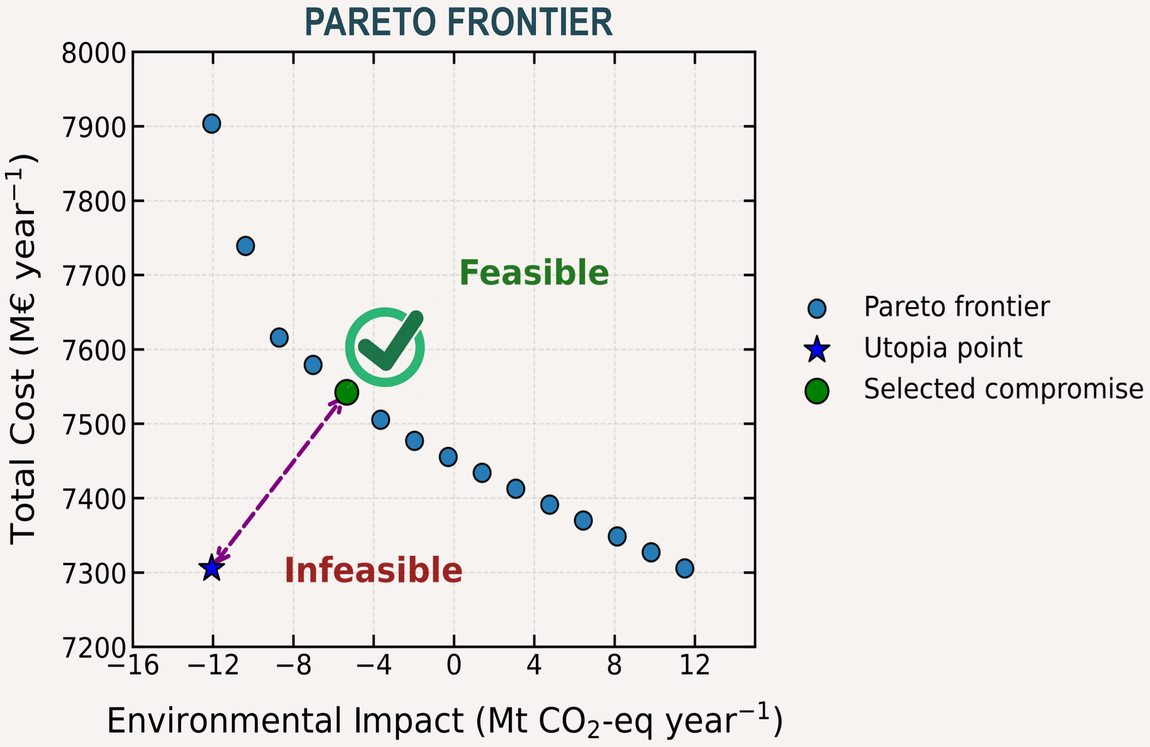
-Selects the best compromise configuration based on the **minimum normalized Euclidean distance to the Utopia (ideal) point**.

3. Life-Cycle and Economic Assessment:

-Evaluates the total system cost and net GHG emissions, accounting for technology conversion, biomass supply and logistics, energy requirements, product substitution and CCS contributions.

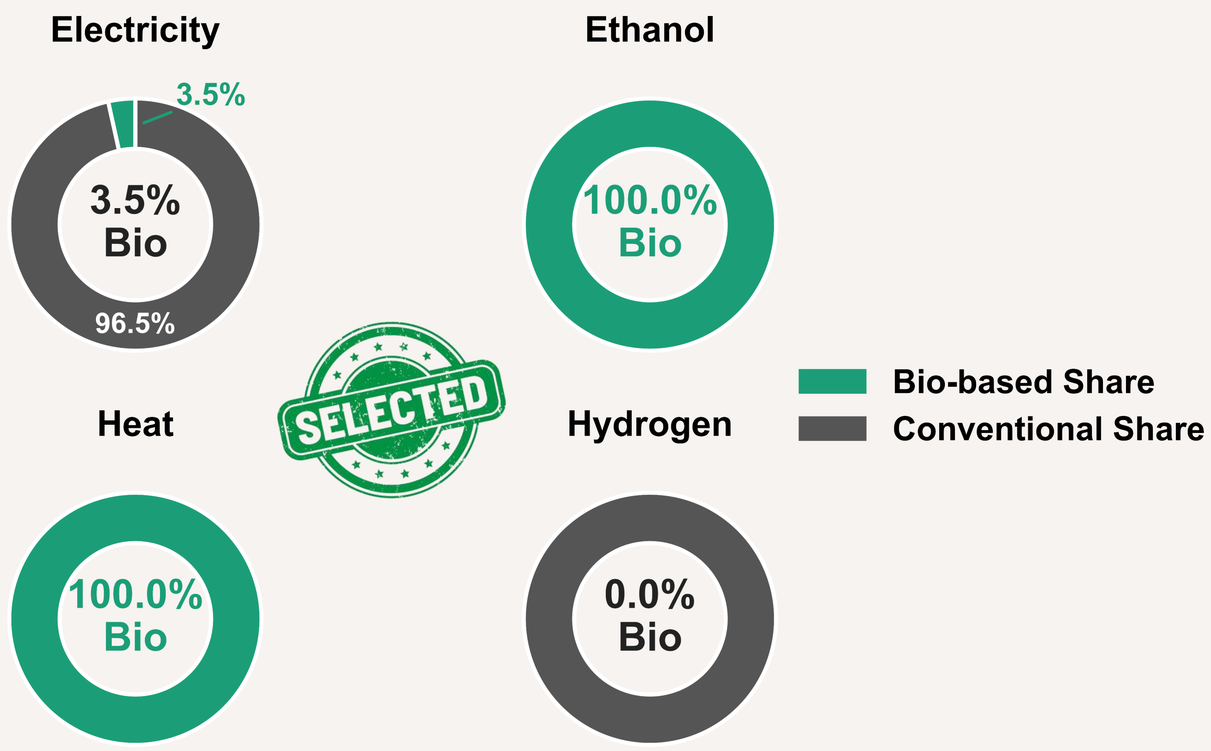
RESULTS AND DISCUSSION

PARETO FRONTIER



- The resulting Pareto front illustrates the trade-off between economic and environmental objectives across alternative network configurations.
- The cost-optimal solution favors conventional routes (**7,306 M€ year^{-1}**) but generates **+11.5 Mt CO2-eq year^{-1}**, whereas the environmental extreme achieves net-negative emissions (**-12.1 Mt CO2-eq year^{-1}**) at an additional cost of **598 M€ year^{-1}**.
- Net-negative emissions can be achieved with a modest increase in total system cost.

DEMAND FULFILLMENT FOR DUAL-ROUTE PRODUCTS



- The selected Utopia-based configuration **reduces from +11.5 to approximately -5.5 Mt CO2-eq year^{-1}** (achieving net-negative emissions) with only a marginal **3.24%** increase in total cost.
- The compromise point prioritizes bio-based pathways (including routes with carbon capture) for high-impact products (100% bio-ethanol and bio-heat), while relying on conventional electricity and hydrogen to prevent cost escalation and achieve an optimal mitigation-cost balance.

CONCLUSIONS

- Economic and environmental objectives lead to markedly different regional biorefinery configurations.
- A small cost increase (+3.24%) enables the transition to a net-negative GHG system in the selected compromise solution.
- Selective deployment of bio-based and CCS-integrated pathways provides a more efficient mitigation strategy than complete substitution of conventional supply.

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